

[Hermes Architecture](#)

# Hermes Agent Architecture

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Public-safe architecture snapshot generated at 2026-06-23T17:06:25-04:00.

Source of truth: local Hermes configuration and runtime status on the operator Linux host.

Sensitive values, private chat IDs, OAuth headers, personal finance/client workflow names, and tokens are intentionally omitted or grouped.

## Executive Summary

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Hermes Agent runs as a **Linux-primary, multi-platform AI operations hub**. The gateway is a systemd service that connects Telegram, Mattermost, WhatsApp, API Server, Home Assistant, MCP servers, cron jobs, skills, memory/context stores, and remote Windows operations.

The default model remains **openai-codex / gpt-5.5**. Local/experimental providers such as **LM Studio** and **Free Kimi** are configured as optional providers only; they are not the default route for Telegram or production workflows.

## High-Level Architecture

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[!Hermes Architecture](#)

## Runtime Topology

| Layer                | Components                                                                                                  | Notes                                                                                                              |
|----------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| User channels        | Telegram, Mattermost, WhatsApp, API Server                                                                  | Gateway adapters route inbound messages into the same Hermes agent core.                                           |
| Agent core           | Hermes Agent gateway + CLI + tool loop                                                                      | Runs on Linux under <code>hermes-gateway.service</code> .                                                          |
| Models               | OpenAI Codex primary, Copilot fallback, optional LM Studio, optional Free Kimi proxy                        | Provider selection is explicit; experimental endpoints are not default.                                            |
| Tools                | Terminal, files, browser, web, vision, TTS, cron, delegation, Home Assistant, GBrain, NotebookLM, CodeGraph | Enabled toolsets differ by platform/session but the gateway exposes the main operational surface.                  |
| Knowledge            | GBrain, NotebookLM, session search, imported Claude context, skills                                         | Long-term memory and research context are split across structured stores and markdown skills.                      |
| Automation           | Hermes cron scheduler, no-agent scripts, hooks, plugins                                                     | Backups, monitoring, GitHub publication, finance snapshots, smart-home logging, news/media digests, and watchdogs. |
| Operational surfaces | Tasks, skills, hooks, plugins, MCP servers, toolsets                                                        | Documented as public-safe counts/categories plus selected non-sensitive examples.                                  |
| Remote systems       | Windows workstation over SSH, Home Assistant, Google Drive/GitHub, local LM Studio                          | Linux remains the control plane; Windows is operated remotely when needed.                                         |

## Operational Surface Inventory

| Surface                | Detected public-safe state                                                      | Notes                                                           |
|------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Scheduled tasks / cron | 35 jobs; 25 no-agent script jobs; 0 agent-backed jobs                           | Exact private task names are grouped by category.               |
| Skills                 | 218 detected skill files across 22 categories                                   | Private/client-sensitive skill names are omitted from examples. |
| Hooks / webhooks       | shell allowlist present: False; allowlist entries: 0; plugin hook manifests: 23 | Hook command bodies are not published.                          |
| Plugins                | 72 visible plugin rows captured; enabled estimate 3                             | Descriptions omitted to avoid leaking credential/env surfaces.  |
| MCP servers            | 6 configured MCP servers                                                        | GBrain, NotebookLM, CodeGraph are the active core MCP surfaces. |

Scheduled tasks / cron categories

| Category                   | Active jobs | Public-safe purpose                                                   |
|----------------------------|-------------|-----------------------------------------------------------------------|
| Backup & sync              | 5           | Protect configuration, repositories, databases, and knowledge stores. |
| GitHub & publishing        | 6           | Maintain GitHub/publication surfaces and repo health digests.         |
| Home automation            | 2           | Log smart-home/home-environment telemetry.                            |
| Knowledge & memory         | 4           | Keep GBrain/memory/context stores healthy and up to date.             |
| Media/news monitoring      | 3           | News, RSS, YouTube, and briefing pipelines.                           |
| Other scheduled automation | 4           | Other local automation jobs.                                          |
| Private finance automation | 4           | Private finance workflow snapshots; details omitted from public docs. |
| Reliability watchdogs      | 7           | Auto-healing, environment guards, timeout/watchdog checks.            |

Skills surface

Hermes currently has a broad skill surface. The public inventory lists category counts and non-sensitive examples only.

| Skill category       | Count |
|----------------------|-------|
| .archive             | 11    |
| apple                | 5     |
| autonomous-ai-agents | 7     |
| creative             | 34    |
| data-science         | 2     |
| devops               | 7     |
| ecc-imports          | 4     |
| email                | 2     |
| gaming               | 2     |
| github               | 11    |
| mcp                  | 1     |
| media                | 7     |
| mlops                | 19    |
| note-taking          | 4     |
| personal             | 5     |
| productivity         | 20    |
| red-teaming          | 1     |
| research             | 13    |
| smart-home           | 5     |
| social-media         | 2     |
| software-development | 37    |
| uncategorized        | 19    |

Public-safe skill examples:

| Skill                   | Category      | Public-safe description                                                                                                                      |
|-------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| dspy                    | mlops         | DSPy: declarative LM programs, auto-optimize prompts, RAG.                                                                                   |
| fine-tuning-with-trl    | mlops         | TRL: SFT, DPO, PPO, GRPO, reward modeling for LLM RLHF.                                                                                      |
| axolotl                 | mlops         | Axolotl: YAML LLM fine-tuning (LoRA, DPO, GRPO).                                                                                             |
| unsloth                 | mlops         | Unsloth: 2-5x faster LoRA/QLoRA fine-tuning, less VRAM.                                                                                      |
| huggingface-hub         | mlops         | HuggingFace hf CLI: search/download/upload models, datasets.                                                                                 |
| segment-anything-model  | mlops         | SAM: zero-shot image segmentation via points, boxes, masks.                                                                                  |
| weights-and-biases      | mlops         | W&B: log ML experiments, sweeps, model registry, dashboards.                                                                                 |
| evaluating-llms-harness | mlops         | lm-eval-harness: benchmark LLMs (MMLU, GSM8K, etc.).                                                                                         |
| llama-cpp               | mlops         | llama.cpp local GGUF inference + HF Hub model discovery.                                                                                     |
| obliteratus             | mlops         | OBLITERATUS: obliterate LLM refusals (diff-in-means).                                                                                        |
| serving-llms-vllm       | mlops         | vLLM: high-throughput LLM serving, OpenAI API, quantization.                                                                                 |
| outlines                | mlops         | Outlines: structured JSON/regex/Pydantic LLM generation.                                                                                     |
| strategic-reading       | uncategorized | Read a book, article, transcript, or case study through the lens of a specific strategic problem you                                         |
| gif-search              | media         | Search/download GIFs from Tenor via curl + jq.                                                                                               |
| spotify                 | media         | Spotify: play, search, queue, manage playlists and devices.                                                                                  |
| youtube-content         | media         | YouTube transcripts to summaries, threads, blogs.                                                                                            |
| article-enrichment      | uncategorized | Transform raw article text dumps in the brain into structured pages with executive summary, verbatim quotes, key insights, why-it-matters, a |
| llm-wiki                | research      | Karpathy                                                                                                                                     |
| blogwatcher             | research      | Monitor blogs and RSS/Atom feeds via blogwatcher-cli tool.                                                                                   |
| [REDACTED]              | research      | Evaluate whether a third-party service or product is trustworthy, integrable, and compliant enough for automation, purchasing, or review for |
| [REDACTED]              | research      | Operate the academic research workflow from paper discovery through experiment-backed writing and submission.                                |

| Skill             | Category      | Public-safe description                                 |
|-------------------|---------------|---------------------------------------------------------|
| polymarket        | research      | Query Polymarket: markets, prices, orderbooks, history. |
| skillpack-check   | uncategorized | /                                                       |
| himalaya          | email         | Himalaya CLI: IMAP/SMTP email from terminal.            |
| chrome-extensions | uncategorized | >                                                       |

Hooks, webhooks, and plugin hook manifests

Hermes has multiple hook-related surfaces: shell-hook allowlists, webhook subscriptions, and imported Claude/Claude plugin hook manifests. The public repo records only surface/count information, not hook command bodies.

| Hook manifest surface                                                                               |
|-----------------------------------------------------------------------------------------------------|
| .claude/plugins/marketplaces/thedotmack/plugin/hooks/hooks.json                                     |
| .claude/plugins/marketplaces/thedotmack/cursor-hooks/hooks.json                                     |
| .claude/plugins/marketplaces/claude-plugins-official/plugins/learning-output-style/hooks/hooks.json |
| .claude/plugins/marketplaces/claude-plugins-official/plugins/security-guidance/hooks/hooks.json     |
| .claude/plugins/marketplaces/claude-plugins-official/plugins/hookify/hooks/hooks.json               |
| .claude/plugins/marketplaces/claude-plugins-official/plugins/ralph-loop/hooks/hooks.json            |
| [REDACTED]                                                                                          |
| .claude/plugins/marketplaces/claude-code-warp/plugins/warp/hooks/hooks.json                         |
| .claude/plugins/marketplaces/claude-code-plugins/plugins/learning-output-style/hooks/hooks.json     |
| .claude/plugins/marketplaces/claude-code-plugins/plugins/security-guidance/hooks/hooks.json         |
| .claude/plugins/marketplaces/claude-code-plugins/plugins/hookify/hooks/hooks.json                   |
| .claude/plugins/marketplaces/claude-code-plugins/plugins/ralph-wiggum/hooks/hooks.json              |
| [REDACTED]                                                                                          |
| .claude/plugins/marketplaces/ponytail/hooks/hooks.json                                              |
| .claude/plugins/cache/thedotmack/claude-mem/13.2.0/hooks/hooks.json                                 |
| .claude/plugins/cache/thedotmack/claude-mem/13.1.0/hooks/hooks.json                                 |
| .claude/plugins/cache/thedotmack/claude-mem/13.0.1/hooks/hooks.json                                 |
| .claude/plugins/cache/thedotmack/claude-mem/13.0.0/hooks/hooks.json                                 |
| .claude/plugins/cache/thedotmack/claude-mem/13.4.0/hooks/hooks.json                                 |
| .claude/plugins/cache/thedotmack/claude-mem/13.3.0/hooks/hooks.json                                 |

## Plugin surface

| Plugin               | Status      |
|----------------------|-------------|
| browser-browser-use  | not enabled |
| browser-browserbase  | not enabled |
| browser-firecrawl    | not enabled |
| chronos              | not enabled |
| basic                | not enabled |
| nous                 | not enabled |
| self-hosted          | not enabled |
| disk-cleanup         | not enabled |
| google_meet          | not enabled |
| fal                  | not enabled |
| krea                 | not enabled |
| openai               | not enabled |
| openai-codex         | not enabled |
| xai                  | not enabled |
| alibaba-provider     | not enabled |
| anthropic-provider   | not enabled |
| arcee-provider       | not enabled |
| bedrock-provider     | not enabled |
| copilot-provider     | not enabled |
| copilot-acp-provider | not enabled |
| custom-provider      | not enabled |
| deepseek-provider    | not enabled |
| gemini-provider      | not enabled |
| gmi-provider         | not enabled |
| huggingface-provider | not enabled |



| Plugin               | Status      |
|----------------------|-------------|
| kilocode-provider    | not enabled |
| kimi-coding-provider | not enabled |
| minimax-provider     | not enabled |
| nous-provider        | not enabled |
| novita-provider      | not enabled |

## Low-Level Surface Files

The repository includes dedicated, low-level public-safe files for each operational surface:

| File                                                | Contents                                                                             |
|-----------------------------------------------------|--------------------------------------------------------------------------------------|
| <a href="#">docs/surfaces/tasks.md</a>              | Scheduled tasks/cron jobs, modes, scripts, schedules, delivery class, workdir class. |
| <a href="#">docs/surfaces/skills.md</a>             | Skill categories, counts, public-safe examples, operational semantics.               |
| <a href="#">docs/surfaces/hooks.md</a>              | Shell hook allowlist state, webhook state, plugin hook manifests.                    |
| <a href="#">docs/surfaces/plugins.md</a>            | Hermes plugin registry rows and status.                                              |
| <a href="#">docs/surfaces/mcp-and-toolsets.md</a>   | MCP servers and toolset count estimates.                                             |
| <a href="#">docs/surfaces/models-and-gateway.md</a> | Model routing, gateway status, channel/platform surface.                             |

## Model Routing

| Role              | Provider       | Model                       | Notes                                         |
|-------------------|----------------|-----------------------------|-----------------------------------------------|
| Primary           | openai-codex   | gpt-5.5                     | Default for Telegram/API/CLI gateway sessions |
| Fallback          | copilot        | gpt-5.4                     | Used when primary fails                       |
| Optional provider | lmstudio       | gemma4unc                   | http://127.0.0.1:1234/v1                      |
| Optional provider | nvidia         | meta/llama-3.3-70b-instruct | https://integrate.api.nvidia.com/v1           |
| Optional provider | freetimi       | cfbt-kimi                   | http://127.0.0.1:3271/v1                      |
| Optional provider | forge_freetimi | cfbt-kimi                   | http://127.0.0.1:8081/v1                      |
| Optional provider | forge_lmstudio | gemma4unc                   | http://127.0.0.1:8082/v1                      |

Local model trial status

| Item               | Status                                                                                               |
|--------------------|------------------------------------------------------------------------------------------------------|
| LM Studio endpoint | available at http://127.0.0.1:1234/v1                                                                |
| Reported model IDs | qwythos9bq5, mythosnanoq6, gemma4coderq3, gemma4coderq4, oymuncq4, [REDACTED], gemma4unc, [REDACTED] |
| Chat smoke test    | `blocked_or_unavailable: {                                                                           |

"error": { "message": "Failed to load model \"gemma4unc\". Error: Model loading was stopped due to insufficient system resources. Under the current settings, this model requires approximately 14.36 GB of memory, and continuing | | Safety decision | Main Hermes remains openai-codex/gpt-5.5`; local provider is optional until a model can load reliably. |

MCP and External Tooling

| MCP server | Transport                            | Public-safe purpose                                                |
|------------|--------------------------------------|--------------------------------------------------------------------|
| gbrain     | Local HTTP MCP on 127.0.0.1:3131/mcp | Personal knowledge graph, memory, code graph, facts, schema tools. |
| notebooklm | npx notebooklm-mcp@latest            | Grounded research over registered Google NotebookLM notebooks.     |
| codegraph  | Local Node command                   | Code intelligence over selected repositories.                      |

## Scheduled Automation

| Category                   | Active jobs | Public-safe purpose                                                   |
|----------------------------|-------------|-----------------------------------------------------------------------|
| Backup & sync              | 5           | Protect configuration, repositories, databases, and knowledge stores. |
| GitHub & publishing        | 6           | Maintain GitHub/publication surfaces and repo health digests.         |
| Home automation            | 2           | Log smart-home/home-environment telemetry.                            |
| Knowledge & memory         | 4           | Keep GBrain/memory/context stores healthy and up to date.             |
| Media/news monitoring      | 3           | News, RSS, YouTube, and briefing pipelines.                           |
| Other scheduled automation | 4           | Other local automation jobs.                                          |
| Private finance automation | 4           | Private finance workflow snapshots; details omitted from public docs. |
| Reliability watchdogs      | 7           | Auto-healing, environment guards, timeout/watchdog checks.            |

## Current Profiles

The live system currently exposes the default profile publicly as:

| Profile  | Model   | Gateway | Alias | Distribution |
|----------|---------|---------|-------|--------------|
| ◆default | gpt-5.5 | running | —     | —            |

Future recommended profile split:

| Profile        | Purpose                                                  | Model stance                                             |
|----------------|----------------------------------------------------------|----------------------------------------------------------|
| default        | Main Telegram/personal assistant                         | openai-codex/gpt-5.5 primary, Copilot fallback.          |
| local-lmstudio | Experimental local-model testing                         | LM Studio only after a local model can pass smoke tests. |
| dev            | Developer workflows with larger context and coding tools | Can selectively use cloud or local providers.            |
| voice-coach    | Low-latency full-duplex voice experiments                | Minimal tools/skills, local STT/TTS first.               |

## Reliability and Safety Boundaries

- Public repo does **not** publish secrets, chat IDs, API keys, OAuth tokens, cookies, private finance details, or client-specific payloads.
- The gateway is supervised by systemd and multiple watchdog cron jobs.
- GBrain runs through a shared local HTTP MCP server to avoid PGLite lock contention.
- Backups and sync jobs run through cron with Google Drive/GitHub targets.
- Windows is treated as a remote worker controlled via SSH, not as the primary Hermes control plane.
- Experimental providers are explicitly optional and must pass direct API + Hermes smoke tests before being used for real agent work.

## Operational Notes

- Hermes version/status summary:

```
Hermes Agent v0.17.0 (2026.6.19) · upstream 64131bf9
Project: ~/.hermes/hermes-agent
Python: 3.11.15
OpenAI SDK: 2.24.0
Update available: 17 commits behind — run 'hermes update'
```

- Fallback chain:

```
Primary:  gpt-5.5  (via openai-codex)

Fallback chain (1 entry):
1. gpt-5.4  (via copilot)

Tried in order when the primary fails (rate-limit, 5xx, connection errors).
Docs: https://hermes-agent.nousresearch.com/docs/user-guide/features/fallback-providers
```

- MCP list:

MCP Servers:

| Name      | Transport             | Tools | Status  |
|-----------|-----------------------|-------|---------|
| codegraph | ~/.nvm/versions/no... | all   | enabled |

|             |                              |             |         |
|-------------|------------------------------|-------------|---------|
| gbrain      | http://127.0.0.1:3131/mcp    | all         | enabled |
| notebooklm  | npx -y notebooklm-mcp@latest | all         | enabled |
| homeway     | https://homeway.io/api/mcp   | all         | enabled |
| windows-cua | ~/.local/bin/windo...        | all         | enabled |
| monarch     | https://api.monarch.com/mcp  | 17 selected | enabled |

## Maintenance

This repository is regenerated by `scripts/update_architecture.py`. The updater is designed to be safe for a public repository:

1. collect local Hermes runtime state,
2. redact secrets and private identifiers,
3. group sensitive scheduled jobs by category,
4. regenerate Markdown, SVG, HTML, JSON inventory, and PDF,
5. commit and push only if files changed.

## Source Files

- `ARCHITECTURE.md` - canonical public architecture document.
- `docs/hermes-architecture.svg` - standalone diagram.
- `docs/hermes-architecture.html` - browser-friendly diagram page.
- `docs/Hermes-Architecture.pdf` - rendered PDF copy.
- `docs/surfaces/*.md` - low-level public-safe inventories for tasks, skills, hooks, plugins, MCP/toolsets, and gateway/model routing.
- `data/inventory.public.json` - redacted machine-readable snapshot.
- `scripts/update_architecture.py` - public-safe regeneration script.

Rendered from GBrain markdown via Hermes brain-pdf.